Newton–Schulz Dynamics Beyond Norm Bounds: Full Spectral Basins, Jordan Rates, and Moore–Penrose Compatibility

HCS-C317 theorem-and-evidence package

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Abstract

For $X_{k+1} = X_k(2I - AX_k)$ we classify exact-arithmetic convergence without replacing spectral radius by a sufficient norm bound. In the invertible case, the basin is exactly $\rho(I - AX_0) < 1$, with a sharp Jordan rate and a complete unit-circle boundary. For arbitrary rectangular A, convergence to $A^\dagger$ holds if and only if the initial matrix has Moore–Penrose support and its range-compressed residual lies in the open spectral disk. The canonical start αA^* is resolved on the open corridor, both endpoints, every exterior face, repeated maximal singular values, and rank zero. Finite exact matrices audit the algebra but do not replace the arbitrary-dimensional proof.

1 Exact residual dynamics

Fix $A \in \mathbb{C}^{m\times n}$ and $X_0 \in \mathbb{C}^{n\times m}$, and define

$$X_{k+1} = X_k(2I_m - AX_k). \tag{1}$$

This reciprocal-matrix iteration goes back to Schulz [1] and Hotelling [2]; generalized-inverse iterations were studied by Ben-Israel [3].

Theorem 1 (Full square basin)**.** *Suppose $m = n$ and A is invertible. With $R_0 = I - AX_0$,*

$$R_k = R_0^{2^k}, \qquad X_k = A^{-1}(I - R_0^{2^k}). \tag{2}$$

Thus $X_k \to A^{-1}$ if and only if $\rho(R_0) < 1$. Whenever $\rho = \rho(R_0) > 0$, let s_* be the largest Jordan-block size among eigenvalues of modulus ρ. Then, for every matrix norm,

$$\|R_k\| = \Theta\Big((2^k)^{s_*-1}\rho^{2^k}\Big). \tag{3}$$

For $\rho = 0$ there is finite termination once 2^k reaches the nilpotency index. For $\rho = 1$, semisimple peripheral blocks give a bounded, nonvanishing residual, whereas a defective peripheral block of maximal size $s_* > 1$ gives $\Theta((2^k)^{s_*-1})$ growth. For $\rho > 1$, (3) is polynomial-times-double-exponential divergence.

Proof. Multiplying (1) by A gives $I - AX_{k+1} = (I - AX_k)^2$, proving (2). Matrix powers tend to zero exactly when their spectral radius is below one. For a Jordan block $J = \lambda I + N$ of size s and an integer $\ell \geq 0$,

$$J^\ell = \sum_{j=0}^{\min(s-1,\ell)} \binom{\ell}{j} \lambda^{\ell-j} N^j. \tag{4}$$

At $\ell = 2^k$, the largest nonzero peripheral term has order $\ell^{s_*-1}\rho^\ell$; equivalence of finite-dimensional norms transfers the two-sided estimate through a fixed similarity. Formula (4) also gives every boundary statement. In particular, "bounded and nonvanishing" at a semisimple unit-circle block does not assert that the residual sequence itself fails to converge. ☐

Revision certificate: full square basin and Jordan boundary

This original round proves residual squaring, the spectral-radius iff basin, and every nilpotent, unit-circle, and exterior Jordan regime.

References

[1] G. Schulz, "Iterative Berechnung der reziproken Matrix," *ZAMM* 13 (1933), DOI: 10.1002/zamm.19330130111.

[2] H. Hotelling, "Some new methods in matrix calculation," *Ann. Math. Stat.* 14 (1943), DOI: 10.1214/aoms/1177731489.

[3] A. Ben-Israel, "An iterative method for computing the generalized inverse of an arbitrary matrix," *Math. Comp.* 19 (1965), DOI: 10.1090/S0025-5718-1965-0179915-5.